

6.5 Electrical Characteristics

(T_A = 25°C, V_{CC} = 5 V to 15 V, unless otherwise specified)⁽¹⁾⁽²⁾

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Supply Voltage		4.5		16	V
Supply Current	V _{CC} = 5 V, R _L = ∞		3	6	mA
	V _{CC} = 15 V, R _L = ∞ (Low State) ⁽³⁾		10	15	
Timing Error, Monostable					
Initial Accuracy			1 %		
Drift with Temperature	R _A = 1 k to 100 kΩ,		50		ppm/°C
	C = 0.1 μF, ⁽⁴⁾				
Accuracy over Temperature			1.5 %		
Drift with Supply			0.1 %		V
Timing Error, Astable					
Initial Accuracy			2.25		
Drift with Temperature	R _A , R _B = 1 k to 100 kΩ,		150		ppm/°C
	C = 0.1 μF, ⁽⁴⁾				
Accuracy over Temperature			3.0%		
Drift with Supply			0.30 %		/V
Threshold Voltage			0.667		x V _{CC}
Trigger Voltage	V _{CC} = 15 V		5		V
	V _{CC} = 5 V		1.67		V
Trigger Current			0.5	0.9	μA
Reset Voltage		0.4	0.5	1	V
Reset Current			0.1	0.4	mA
Threshold Current	⁽⁵⁾		0.1	0.25	μA
Control Voltage Level	V _{CC} = 15 V	9	10	11	V
	V _{CC} = 5 V	2.6	3.33	4	
Pin 7 Leakage Output High			1	100	nA
Pin 7 Sat ⁽⁶⁾					
Output Low	V _{CC} = 15 V, I _T = 15 mA		180		mV
Output Low	V _{CC} = 4.5 V, I _T = 4.5 mA		80	200	mV
Output Voltage Drop (Low)	V _{CC} = 15 V				
	I _{SINK} = 10 mA		0.1	0.25	V
	I _{SINK} = 50 mA		0.4	0.75	V
	I _{SINK} = 100 mA		2	2.5	V
	I _{SINK} = 200 mA		2.5		V
	V _{CC} = 5 V				
	I _{SINK} = 8 mA				V
	I _{SINK} = 5 mA		0.25	0.35	V

(1) All voltages are measured with respect to the ground pin, unless otherwise specified.

(2) **Absolute Maximum Ratings** indicate limits beyond which damage to the device may occur. **Recommended Operating Conditions** indicate conditions for which the device is functional, but do not ensure specific performance limits. **Electrical Characteristics** state DC and AC electrical specifications under particular test conditions which ensures specific performance limits. This assumes that the device is within the **Recommended Operating Conditions**. Specifications are not ensured for parameters where no limit is given, however, the typical value is a good indication of device performance.

(3) Supply current when output high typically 1 mA less at V_{CC} = 5 V.

(4) Tested at V_{CC} = 5 V and V_{CC} = 15 V.

(5) This will determine the maximum value of R_A + R_B for 15 V operation. The maximum total (R_A + R_B) is 20 MΩ.

(6) No protection against excessive pin 7 current is necessary providing the package dissipation rating will not be exceeded.

Electrical Characteristics (continued)

($T_A = 25^\circ\text{C}$, $V_{CC} = 5\text{ V}$ to 15 V , unless otherwise specified)⁽¹⁾⁽²⁾

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Output Voltage Drop (High)	$I_{\text{SOURCE}} = 200\text{ mA}$, $V_{CC} = 15\text{ V}$		12.5		V
	$I_{\text{SOURCE}} = 100\text{ mA}$, $V_{CC} = 15\text{ V}$	12.75	13.3		V
	$V_{CC} = 5\text{ V}$	2.75	3.3		V
Rise Time of Output			100		ns
Fall Time of Output			100		ns

6.6 Typical Characteristics

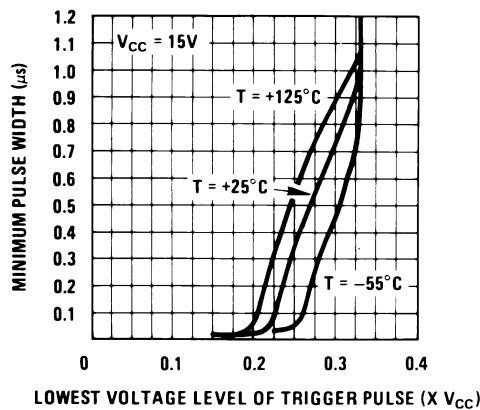


Figure 1. Minimum Pulse Width Required For Triggering

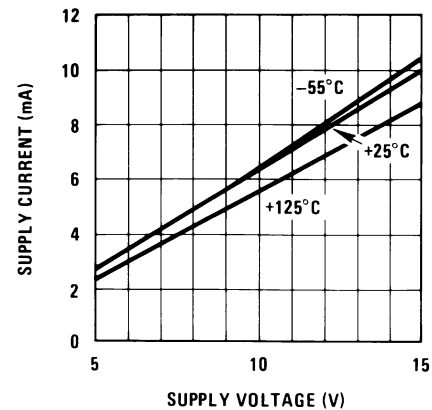


Figure 2. Supply Current vs. Supply Voltage

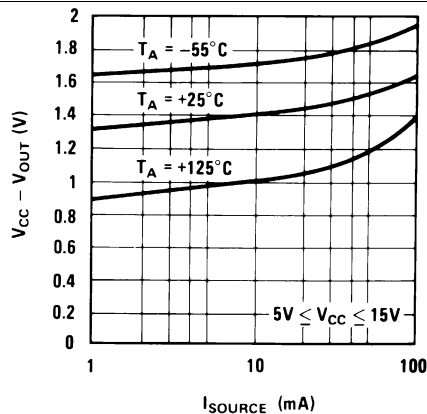


Figure 3. High Output Voltage vs. Output Source Current

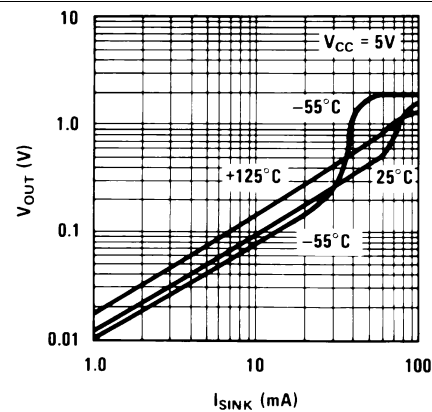


Figure 4. Low Output Voltage vs. Output Sink Current